

L2 Development Guide



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L2 Development Guide

Preface

- L2 is only a LiDAR sensor and does not have intelligent analysis and data storage functions. The application of actual scenarios requires secondary development
- After L2 is powered on and works normally, it will continuously send point cloud IMU and other data measured by the LiDAR
- The Unilidar 2 host computer provided by L2 is only suitable for simple testing experience. It only has basic point cloud display and parameter configuration functions, and does not have data processing, storage and export functions
- L2 time synchronization needs to rely on timestamps. The moment after the radar is fully started is the starting point. Each packet of L2 data will be timestamped. The SDK contains the system time synchronized to the radar. Please pay attention to the errors caused by network delays and other reasons when using it.
- L2's Unilidar SDK 2 provides the following common interfaces: ① [Original C++-based SDK package](#) ② [ROS environment package](#) ③ [ROS2 environment package](#). For detailed usage, please refer to the corresponding chapters
- When the development environment is Windows or other environments where SDK cannot be used, you need to continue reading the L2 development guide to parse the original data to obtain the point cloud IMU and other data of the laser radar

Glossary

Header	Frame header in protocol packet
Tail	Frame tail in protocol packet
NEGA	Refers to the mode where the single-line angle range of the laser radar is 192°
Normal	Radar working mode
StandBy	Radar standby mode
DataInfo	Data packet information, mainly including sequence number and timestamp

Overview

L2 laser radar supports two data output modes, UDP and UART, to achieve a large angle measurement range, and supports reflectivity detection, dirt detection, software time synchronization function, and comes with IMU attitude data output. L2 laser radar has multiple working modes and can be flexibly configured.

- Supports 3D working mode and 2D working mode: 3D point cloud data with a range of 360x90°/360x96° can be output in 3D; point cloud data with a range of 180°/192° can be output in 2D mode.
- Supports 500Hz IMU data reporting, including raw angular velocity, acceleration data and quaternion.
- Point cloud data supports 255-level grayscale information.

- Supports software time synchronization, configurable timestamp, and proprietary communication package calculation system communication delay function.
- Supports dirt detection. If dust adheres to the radar surface and affects the radar measurement effect, dirt index information will be provided.
- Uses spatial polar coordinates to output point cloud information, providing more point cloud information with a lower data volume.
- Supports StandBy mode switching, in which the motor and IMU do not work.
- Supports IP address port number configuration and MAC address configuration.

Development method

Using the L2 SDK provided by Unitree to develop Unitree 4D LiDAR L2 laser radar is the simplest and fastest choice for most users. The official SDK provides a comprehensive, detailed and accurate solution to help you use L2 laser radar development projects.

For some special customers who need to directly read L2 laser radar data, we also provide the underlying interface of L2 laser radar for users to obtain radar point cloud information and mode configuration. Theoretically, L2 software supports the development and use of any platform and language, and L2 hardware also supports a wide range of application scenarios.

Mode Introduction

L2 LiDAR provides a variety of working modes as follows

- NEGA Enable:
L2 works in normal angle mode by default, with a line scanning angle of 180°. It can also be configured to NEGA mode, increasing the line angle to 192° to meet the needs of users with negative angles.

- 3D/2D Mode:

In 3D mode, the scanning range of the normal angle mode radar is 360x90°, and it is increased to 360x96° in NEGA mode. Similarly, in 2D mode, the normal angle mode has a measurement range of 180°, and the NEGA mode has a measurement range of 192°.

- IMU Enable:

For the high frequency of IMU reporting data and the large amount of data, some users who do not use IMU data can configure L2 to IMU disable mode, and L2 will not output IMU data packets.

- ENET/UART selection:

L2 LiDAR supports two communication modes: ENET and UART. In ENET mode, data is reported via UDP; in UART mode, data is reported via UART serial port.

- Startup mode

L2 supports two startup modes: power-on auto-start mode and power-on wait command startup mode.

In power-on auto-start mode, as long as L2 is connected to a suitable power supply, the radar will start and output point cloud and attitude data.

In command mode, L2 will enter StandBy mode when powered on, and the radar can only be started and enter working mode through communication commands.

- Gray Enable selection:

L2 can be configured to enable and disable reflectivity data. It is recommended to configure the disable reflectivity mode in scenarios where reflectivity data is not required.

All the above working modes can be realized by configuring radar registers, and power-off saving is supported.

Communication protocol details

L2 supports both Ethernet UDP protocol and UART hardware interfaces for communication with the host. The software communication protocol is exactly the same in both modes. The baud rate of the serial communication is 4Mbit/s, with 8 data bits, 1 stop bit, and None parity bit. Due to the data volume of L2 communication, the baud rate here is relatively high. Users need to pay attention to whether their hardware supports this baud rate. It is recommended to use the officially equipped equipment for connection. Note: When using the network port for communication, the tx pin on the serial port PC side should not be left floating, otherwise the radar will output an error response ACK packet.

	IP address	UDP port number
L2	192.168.1.62	6101
Host	192.168.1.2	6201

The communication interfaces between Unitree L2 and the host are mainly as follows. Data packet types are mainly divided into three categories. Parsing data packets require users to parse the data packet content by themselves to obtain the corresponding radar working data and configuration data, etc.; configuration data packets require users to configure the radar working mode, IP address or Mac address, etc., and users need to actively send data to LiDAR. In addition, after the configuration is completed, power must be turned off for use. This type of command is saved after power off; command data packets are also sent by users to the radar to obtain radar information or change the radar working state through commands, such as the switch between Normal mode and StandBy mode, which is implemented in the radar working state. Most commands do not require power off and do not support power off saving.

(Communication Packet) Name	Function	Type	Packet Length	Send Interval
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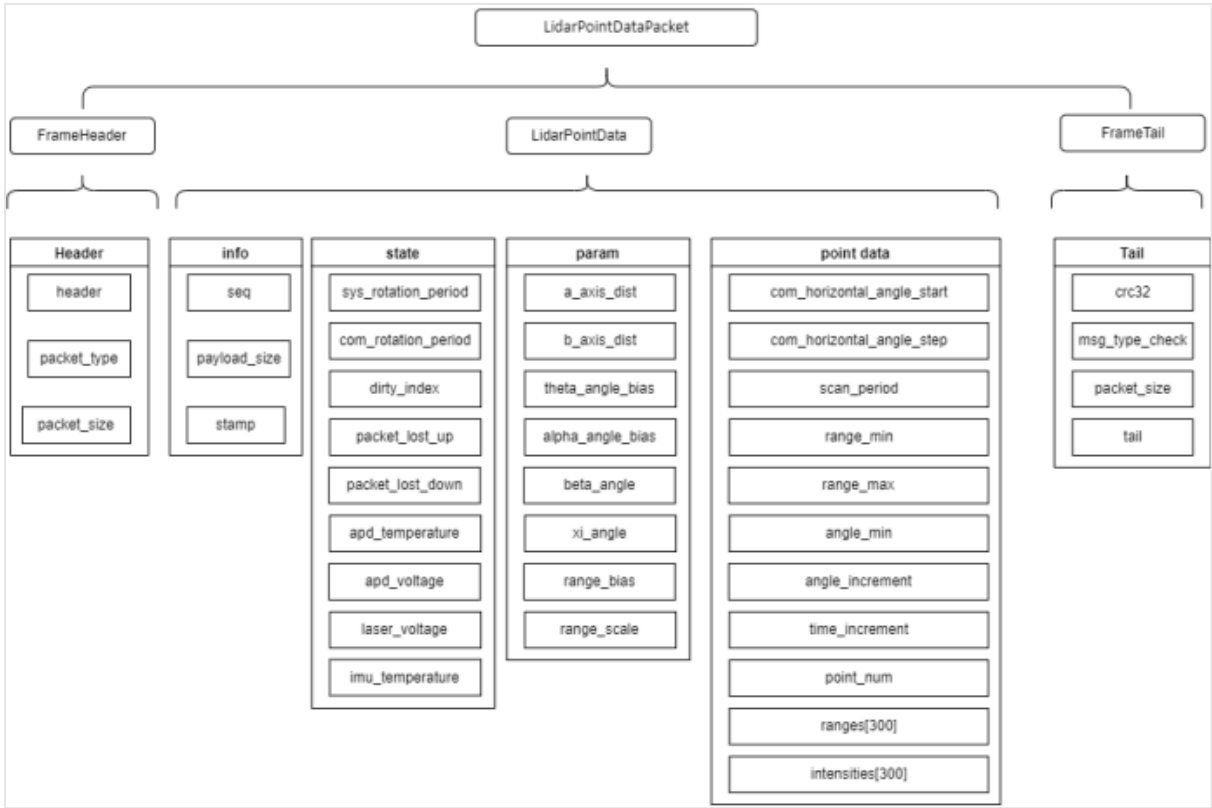
LidarPointDataPacket	3D Point Cloud Data Output	UDP/UART	1044byte	About 4.63 ms
Lidar2DPointDataPacket	2D point cloud data output	UDP/UART	5536byte	About 28ms
LidarImuDataPacket	IMU data output	UDP/UART	80byte	About 2ms
LidarAckDataPacket	Command response	UDP/UART	40byte	INF
LidarVersionDataPacket	Version number packet output	UDP/UART	104byte	INF
LidarIpAddressConfigPacket	IP address configuration packet	UDP/UART	44byte	INF
LidarMacAddressConfigPacket	MAC address configuration packet	UDP/UART	32byte	INF
LidarWorkModeConfigPacket	Work mode configuration packet	UDP/UART	28byte	INF
LidarUserCtrlCmdPacket	User command packet	UDP/UART	32byte	INF

Data packet protocol format

Note: For the specific number of bytes corresponding to each structure, please refer to the definition in `unitree_lidar_sdk\include\unitree_lidar_protocol.h` under the root directory of the sdk software package.

LidarPointDataPacket

The LidarPointDataPacket package mainly outputs 3D point cloud data, including basic information, radar status, dirt index, calibration data, horizontal angle of the starting point, horizontal angle interval of adjacent points, vertical angle of the starting point, vertical angle interval of adjacent points, time interval of adjacent points, distance data unit, number of points in a single package, distance data set, reflectivity data set, etc. The length of LidarPointDataPacket is 1032 bytes, including 12 bytes of Header, 12 bytes of DataInfo, 36 bytes of radar status data, 32 bytes of calibration parameters, 600 bytes of distance data, 300 bytes of reflectivity data, and 12 bytes of Tail.



1. Frame header

The frame header is 12 bytes in total, which is used to identify the data location and type.

The 4 bytes of the Header: field are defined as 0x55, 0xAA, 0x05, 0x0A, which are used to identify the start position of the data.

The packet_type: field is used to indicate the type of the current data packet.

The Packet_size: field is used to indicate the size of the current data packet.

2. Radar point cloud data

The DataInfo structure contains the following fields

Seq: The field is used to indicate the sequence of the current data packet, with a maximum value of 1024.

Payload_size: The field is used to indicate the size of the current point cloud packet.

Stamp: The field contains timestamp information.

The LidarInsideState structure contains the radar status information field

sys_rotation_period: The field is used to indicate the high-speed motor rotation period, in microseconds.

com_rotation_period: The field is used to indicate the low-speed motor rotation period, in microseconds.

dirty_index: The field is used to indicate the dirty index.

packet_lost_up: The field is used to indicate the quality of the optical communication uplink signal.

packet_lost_down: The field is used to indicate the quality of the optical communication downlink signal.

apd_temperature: The field is used to indicate the APD temperature.

apd_voltage: The field is used to indicate the APD voltage.

laser_voltage: The field is used to indicate the laser emission voltage.

imu_temperature: The field is used to indicate the current temperature of the IMU.

The LidarCalibParam structure contains the output point cloud calibration information field, and the content of this field does not need to be paid

attention to. It can be used directly according to the official calculation method or interface function (see the subsequent chapters).

The data related to point cloud display includes the following fields:

`com_horizontal_angle_start`: The horizontal angle of the first point of the data packet, in rad.

`com_horizontal_angle_step`: The horizontal angle difference between adjacent points in the data packet, in rad.

`scan_period`: The data packet period, in s.

`range_min`: The closest measured distance, in m.

`range_max`: The farthest measured distance, in m.

`angle_min`: The angle of the first point in the vertical direction, in rad.

`angle_increment`: The angle difference between adjacent points in the vertical direction, in rad.

`time_increment`: Time difference between adjacent points, unit s.

`point_num`: The number of points in the data package, generally 1800 in 2D mode.

Ranges: Point cloud distance data, unit mm.

Intensities: Point cloud reflectivity data, value range 0~255.

3. Frame tail

`crc32`: CRC check value of `LidarPointData`.

Reserve: uint32 reserve.

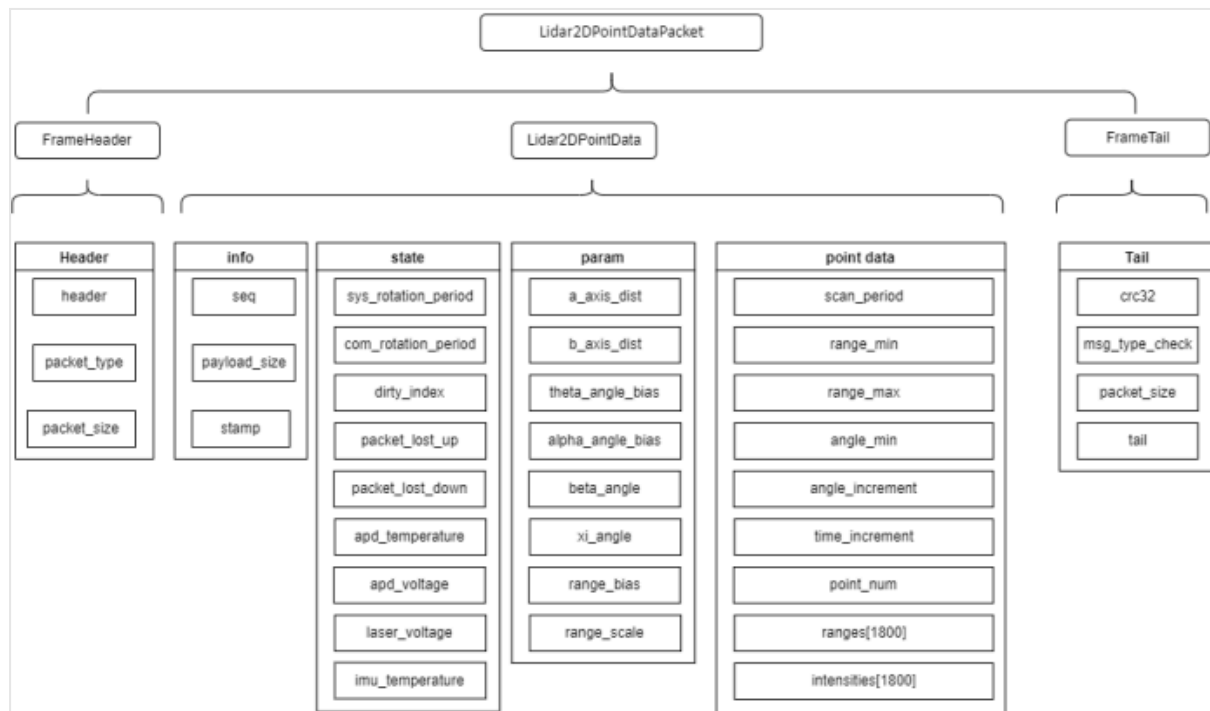
`Resetve[2]`: uint8 resetve.

Tail: Check frame tail, 0x00, 0xFF.

Lidar2DPointDataPacket

The `Lidar2DPointDataPacket` package mainly outputs 2D point cloud data, including basic information, radar status, dirt index, calibration data starting point angle, adjacent point angle interval, adjacent point time interval, distance data unit, number of points in a single package, distance data set, reflectivity data set, etc. The length of the `Lidar2DPointDataPacket` is 5536 bytes, including 12 bytes `FrameHeader`, 5512 bytes of `Lidar2DPointData` main data (radar data, angle information, status

information, etc.), and 12 bytes of FrameTail.



1. Frame Header

The frame header is 12 bytes in total, which is used to identify the data location and type.

The 4 bytes of the Header: field are defined as 0x55, 0xAA, 0x05, 0x0A, which are used to identify the starting position of the data.

The packet_type: field is used to indicate the type of the current data packet.

The Packet_size: field is used to indicate the size of the current data packet.

2. Radar point cloud data

The DataInfo structure contains the following fields

Seq: The field is used to indicate the sequence of the current data packet, with a maximum value of 1024.

Payload_size: The field is used to indicate the size of the current point cloud packet.

Stamp: The field contains timestamp information.

The LidarInsideState structure contains the radar status information field

sys_rotation_period: The field is used to indicate the high-speed motor rotation period, in microseconds.

com_rotation_period: The field is used to indicate the low-speed motor rotation period, in microseconds.

dirty_index: The field is used to indicate the dirty index.

packet_lost_up: The field is used to indicate the quality of the optical communication uplink signal.

packet_lost_down: The field is used to indicate the quality of the optical communication downlink signal.

apd_temperature: The field is used to indicate the APD temperature.

apd_voltage: The field is used to indicate the APD voltage.

laser_voltage: The field is used to indicate the laser emission voltage.

imu_temperature: The field is used to indicate the current temperature of the IMU.

The LidarCalibParam structure contains the output point cloud calibration information field, and the content of this field does not need to be paid attention to. You can use it directly according to the calculation method or interface function provided by the official (see the subsequent chapters).

The data related to point cloud display includes the following fields:

scan_period: data packet period, unit s.

range_min: the closest measured distance, unit m.

range_max: the farthest measured distance, unit m.

angle_min: the angle of the first point in the data packet, unit rad.

angle_increment: the angle difference between adjacent points, unit rad.

time_increment: the time difference between adjacent points, unit s.

point_num[1800]: the number of points in the data packet, usually 1800 in 2D mode.

Ranges[1800]: point cloud distance data, unit mm.

Intensities: point cloud reflectivity data, value range 0~255.

3. Frame end

crc32: crc32 checksum value of LidarPointData.

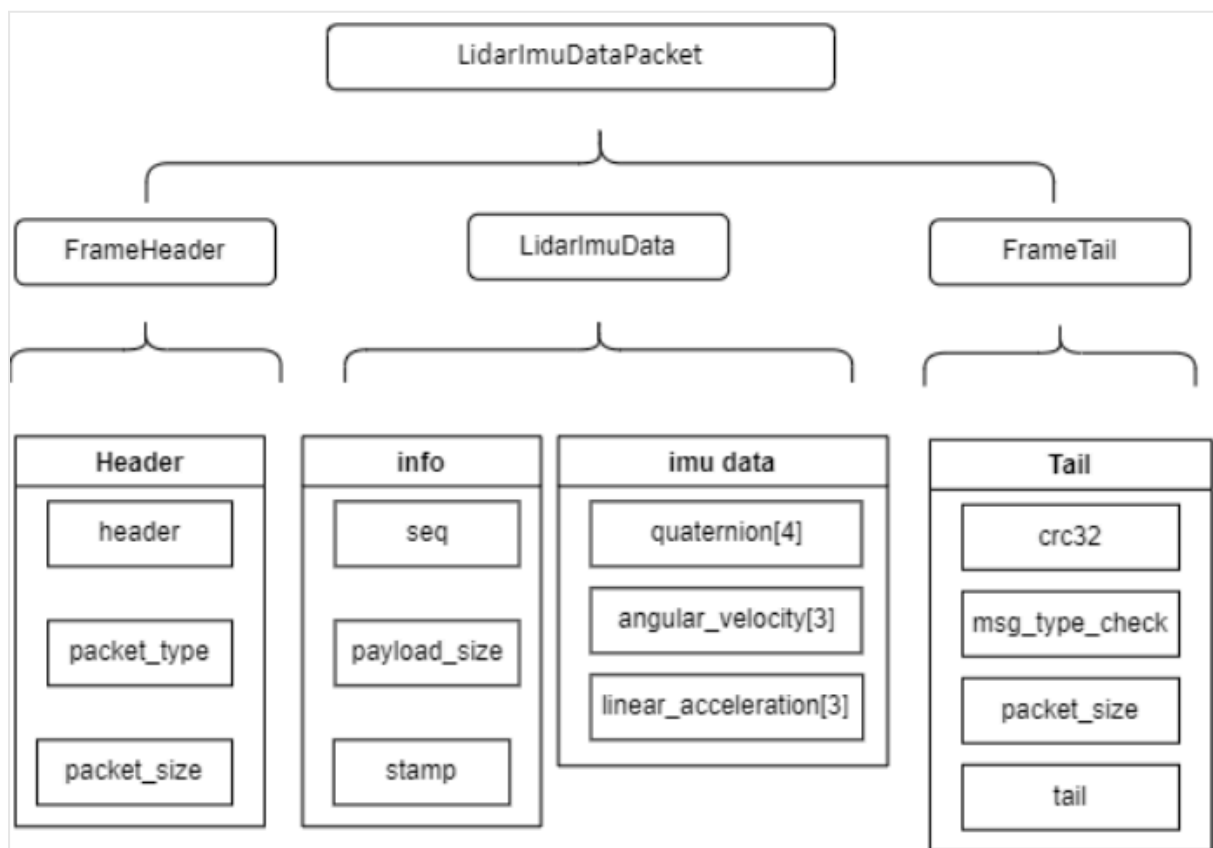
Reserve: uint32 reserve.

reserve [2]: uint8 reserve.

tail[2]: check frame tail, 0x00, 0xFF.

LidarImuDataPacket

The LidarImuDataPacket package mainly outputs IMU attitude information, including basic information, IMU quaternion, IMU acceleration raw value, and angular velocity raw value. The total length of LidarImuDataPacket is 80 bytes, including 12 bytes of FrameHeader, 56 bytes of IMU attitude data LidarImuData, and 12 bytes of FrameTail.



1. Frame Header

The frame header is 12 bytes in total, which is used to identify the data location and type.

Header: The 4 bytes of the field are defined as 0x55, 0xAA, 0x05, 0x0A, which are used to identify the start position of the data.

packet_type: field is used to indicate the type of the current data packet.

Packet_size: field is used to indicate the size of the current data packet.

2. IMU data

The DataInfo structure contains the following fields

Seq: field is used to indicate the sequence of the current data packet, with a maximum value of 1024.

payload_size: field is used to indicate the size of the current point cloud packet.

Stamp: field contains timestamp information.

The LidarImuData structure mainly includes IMU quaternion and acceleration angular velocity raw data.

quaternion[4]: field is used to indicate the quaternion output by the IMU.

angular_velocity[3]: 3-axis angular velocity raw data, measurement range.

linear_acceleration[3]: 3-axis acceleration raw data.

3. Frame tail

crc32: crc32 checksum of LidarPointData.

Reserve: uint32 reserve.

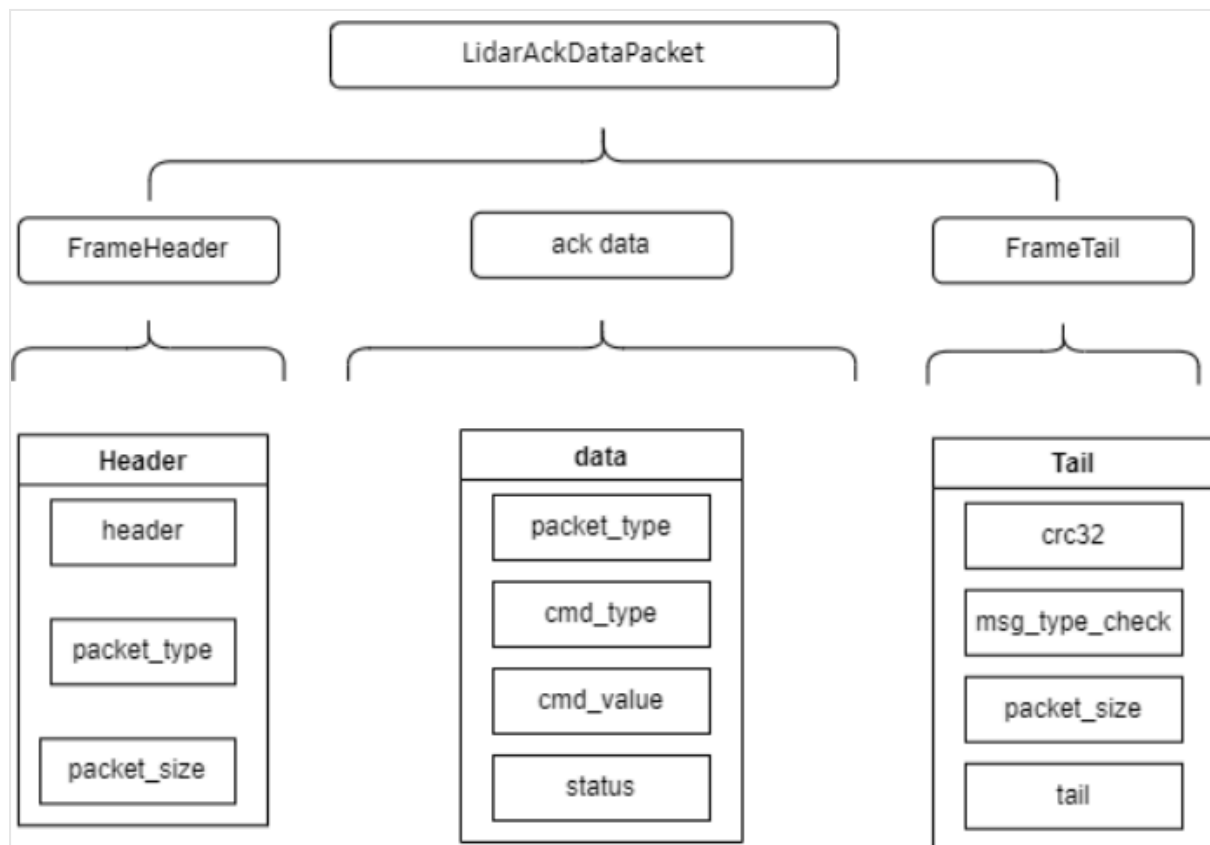
reserve [2]: uint8 reserve.

tail[2]: check frame tail, 0x00, 0xFF.

LidarAckDataPacket

The LidarAckDataPacket package is used to reply to the instruction package sent by the PC. Every time a configuration parameter package, command package, etc. are received, this ACK package must be replied. The total length of the ACK packet is 40

bytes, including a 12-byte header, a 16-byte reply data type, and a 12-byte tail.



1. Frame header

The frame header is 12 bytes in total and is used to identify the data location and type.

The 4 bytes of the Header: field are defined as 0x55, 0xAA, 0x05, 0x0A, which are used to identify the start position of the data.

The **packet_type**: field is used to indicate the type of the current data packet.

The **Packet_size**: field is used to indicate the size of the current data packet.

2. ACK field

The **packet_type** field indicates the data packet type of the current ACK packet.

The **cmd_type** field indicates the data packet type replied by the current ACK packet.

The **cmd_value** field indicates the command value of the current ACK packet.

The **status** field feedbacks the processing exception.

ACK_SUCCESS The communication has been successfully processed.

ACK_CRC_ERROR The communication packet CRC is abnormal.

ACK_HEADER_ERROR The communication packet frame header is abnormal.

ACK_BLOCK_ERROR The communication packet is blocked abnormally.

ACK_WAIT_ERROR The communication packet has been received, but the response data has not been generated yet. The request will be sent again later.

3. Frame tail

crc32: CRC32 checksum value of LidarPointData.

Reserve: uint32 reserve.

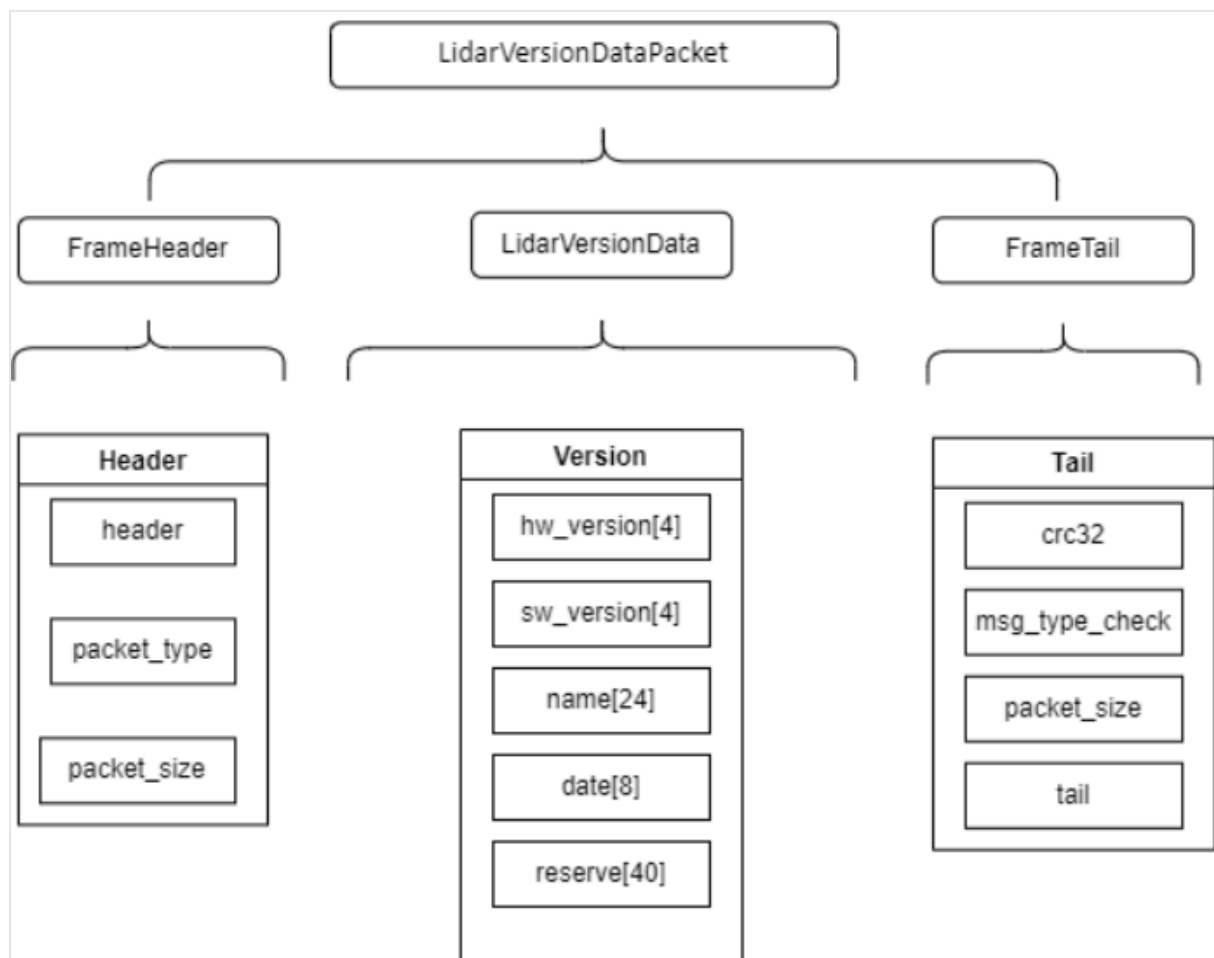
reserve [2]: uint8 reserve.

tail[2]: Check frame tail, 0x00, 0xFF.

LidarVersionDataPacket

LidarVersionDataPacket mainly provides the radar version number, including software version number, hardware version number, date, notes, etc. The packet length is 104 bytes, including 12 bytes of FrameHeader, 80 bytes of version

information LidarVersionData, and 12 bytes of FrameTail.



1. Frame header

The frame header is 12 bytes in total, which is used to identify the data location and type.

Header: The 4 bytes of the field are defined as 0x55, 0xAA, 0x05, 0x0A, which are used to identify the starting position of the data.

packet_type: The field is used to indicate the type of the current data packet.

Packet_size: The field is used to indicate the size of the current data packet.

2. Version field

hw_version[4]: Hardware version number.

sw_version[4]: Software version number.

name[24]: Device name.

date[8]: Compile date.

reserve[40]: Reserve.

3. Frame tail

crc32: CRC32 checksum of LidarPointData.

Reserve: uint32 reserve.

reserve [2]: uint8 reserve.

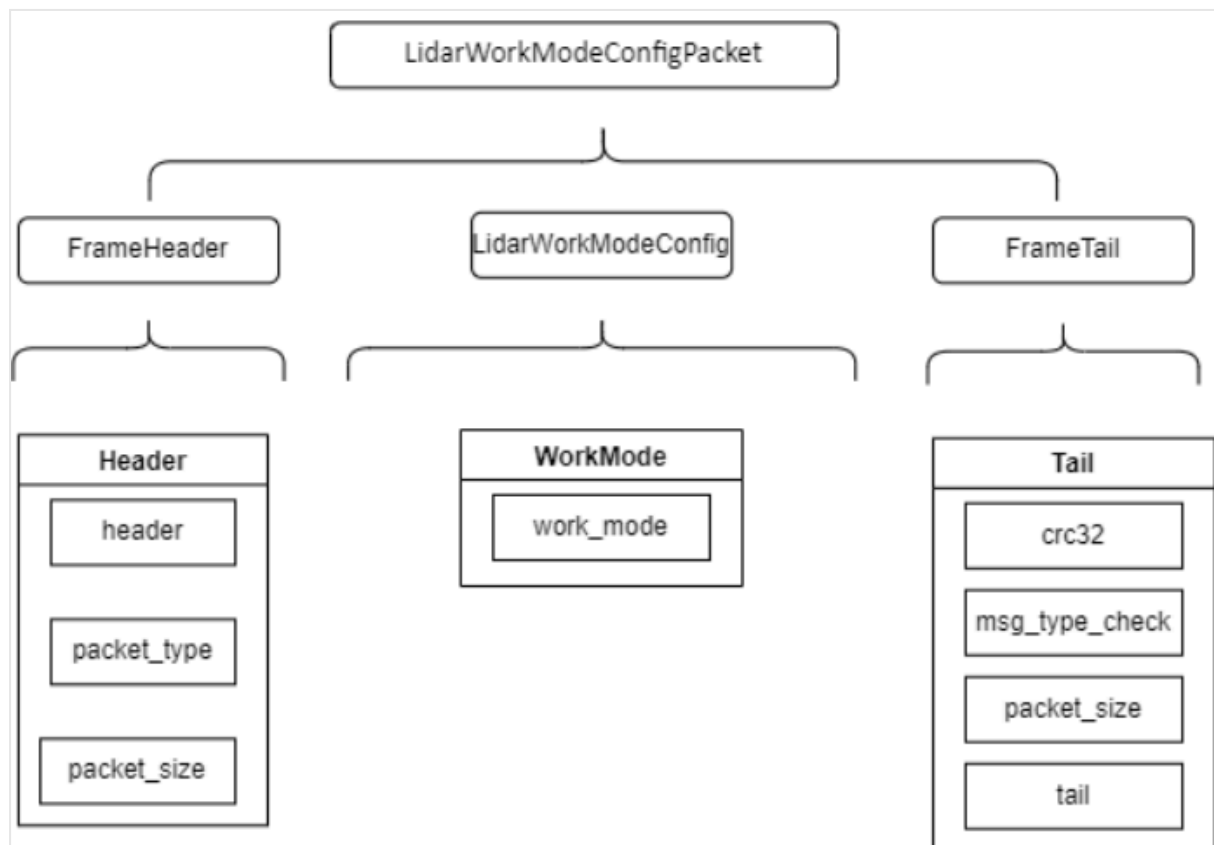
tail[2]: Check frame tail, 0x00, 0xFF.

******The above five data packets need to be parsed and implemented by the user. The following three types of data packets are configuration data packets, which need to be initiated by the PC. ******

LidarWorkModeConfigPacket

The LidarWorkModeConfigPacket mode configuration data packet is used to configure the working mode of the radar. After the configuration is successful, it is saved when the power is turned off. The configured working mode is executed by default at the next startup. The packet length is 32 bytes, with a 12-byte Header, a

4-byte mode configuration field, and a 12-byte Tail.



1. Frame header

The frame header is 12 bytes in total, which is used to identify the data location and type.

The 4 bytes of the Header: field are defined as 0x55, 0xAA, 0x05, 0x0A, which are used to identify the start position of the data.

The packet_type: field is used to indicate the type of the current data packet.

The packet_size: field is used to indicate the size of the current data packet.

2. Work mode field

work_mode: mode configuration register, see the L2 work mode configuration table for details.

3. Frame tail

crc32: crc32 checksum of LidarPointData.

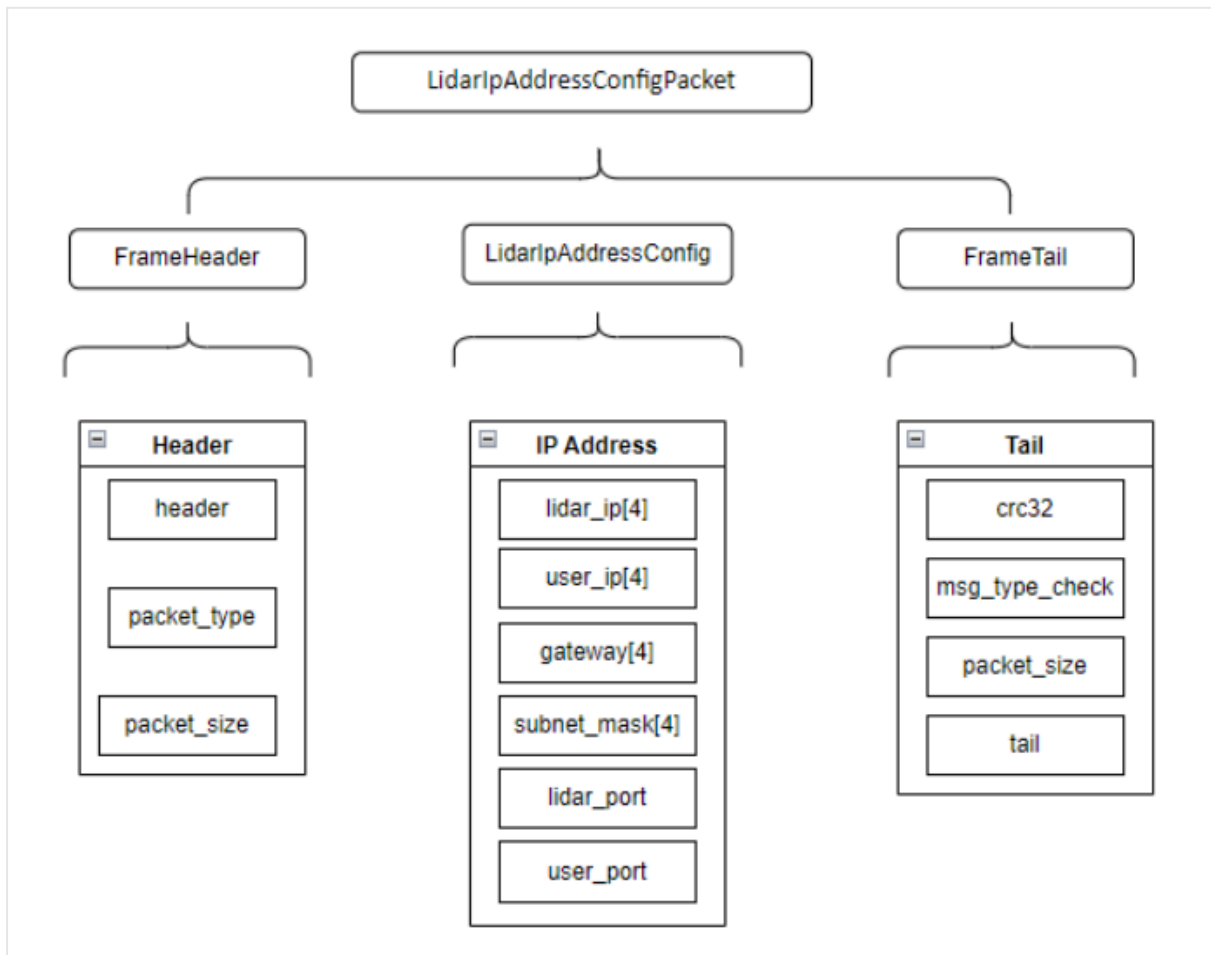
Reserve: uint32 reserve.

reserve [2]: uint8 reserve.

tail[2]: check frame tail, 0x00, 0xFF.

LidarIpAddressConfigPacket

LidarIpAddressConfigPacket can configure IP address and port number. The total length of the data packet is 44 bytes, with 12 bytes of header, 20 bytes of IP field and port field, and 12 bytes of tail.



1. Frame header

The frame header is 12 bytes in total, which is used to identify the data location and type.

Header: The 4 bytes of the field are defined as 0x55, 0xAA, 0x05, 0x0A, which are used to identify the starting position of the data.

packet_type: The field is used to indicate the type of the current data packet.

Packet_size: The field is used to indicate the size of the current data packet.

2. IP address field

lidar_ip[4]: L2 IP address setting field.

user_ip[4]: PC IP address setting field.

gateway: Gateway field.

subnet_mask: Subnet mask.

lidar_port: L2 port number setting field.

user_port: PC port number setting field.

3. Frame tail

crc32: CRC32 checksum of LidarPointData.

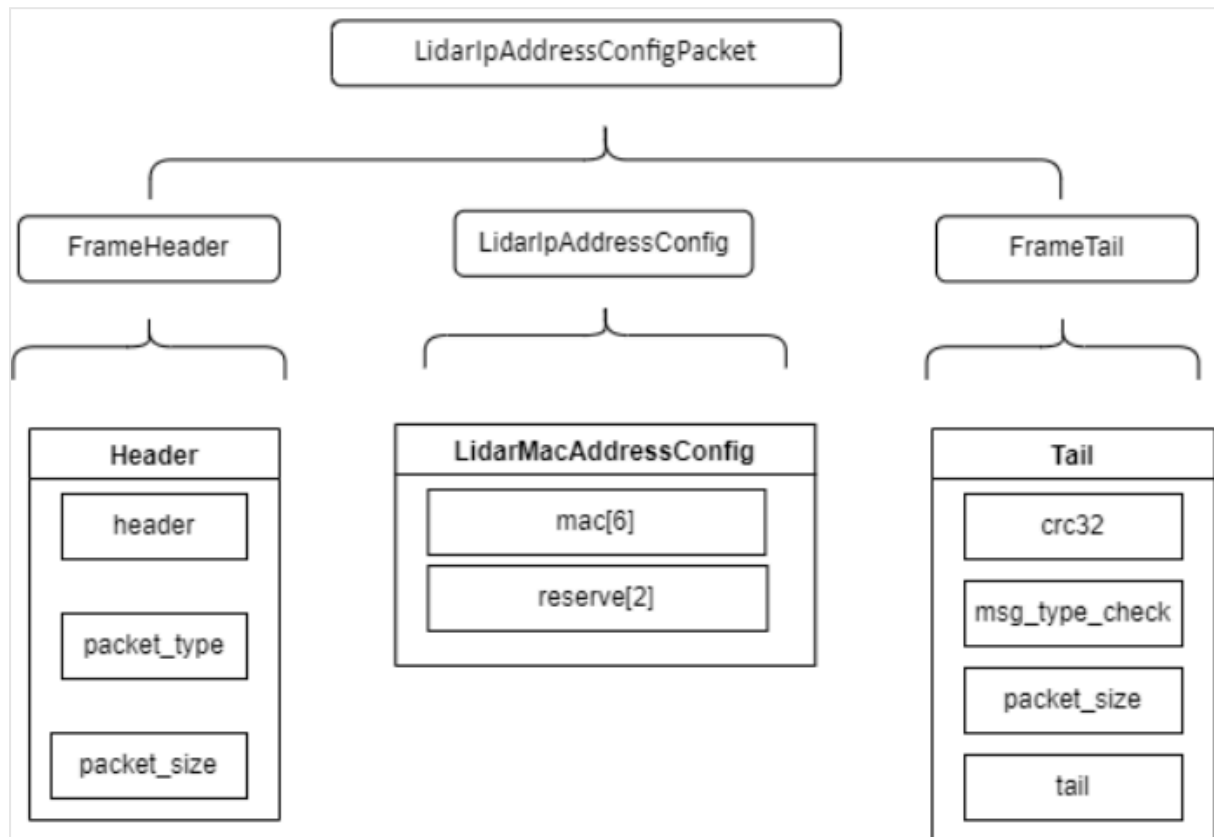
Reserve: uint32 reserve.

reserve [2]: uint8 reserve.

tail[2]: Check frame tail, 0x00, 0xFF.

LidarMacAddressConfigPacket

LidarMacAddressConfigPacket can implement MAC address configuration. The total length of the data packet is 28 bytes, with a 12-byte header, a 6-byte MAC field, a 2-byte reserve field, and a 12-byte tail.



1. Frame header

The frame header is 12 bytes in total, used to identify the data location and

type.

Header: The 4 bytes of the field are defined as 0x55, 0xAA, 0x05, 0x0A, which are used to identify the start position of the data.

packet_type: The field is used to indicate the type of the current data packet.

Packet_size: The field is used to indicate the size of the current data packet.

2. MAC address field

mac[6]: L2 MAC address setting field.

reserve[2]: Reserved field, no need to pay attention.

3. Frame tail

crc32: CRC32 checksum of LidarPointData.

Reserve: uint32 reserve.

reserve [2]: uint8 reserve.

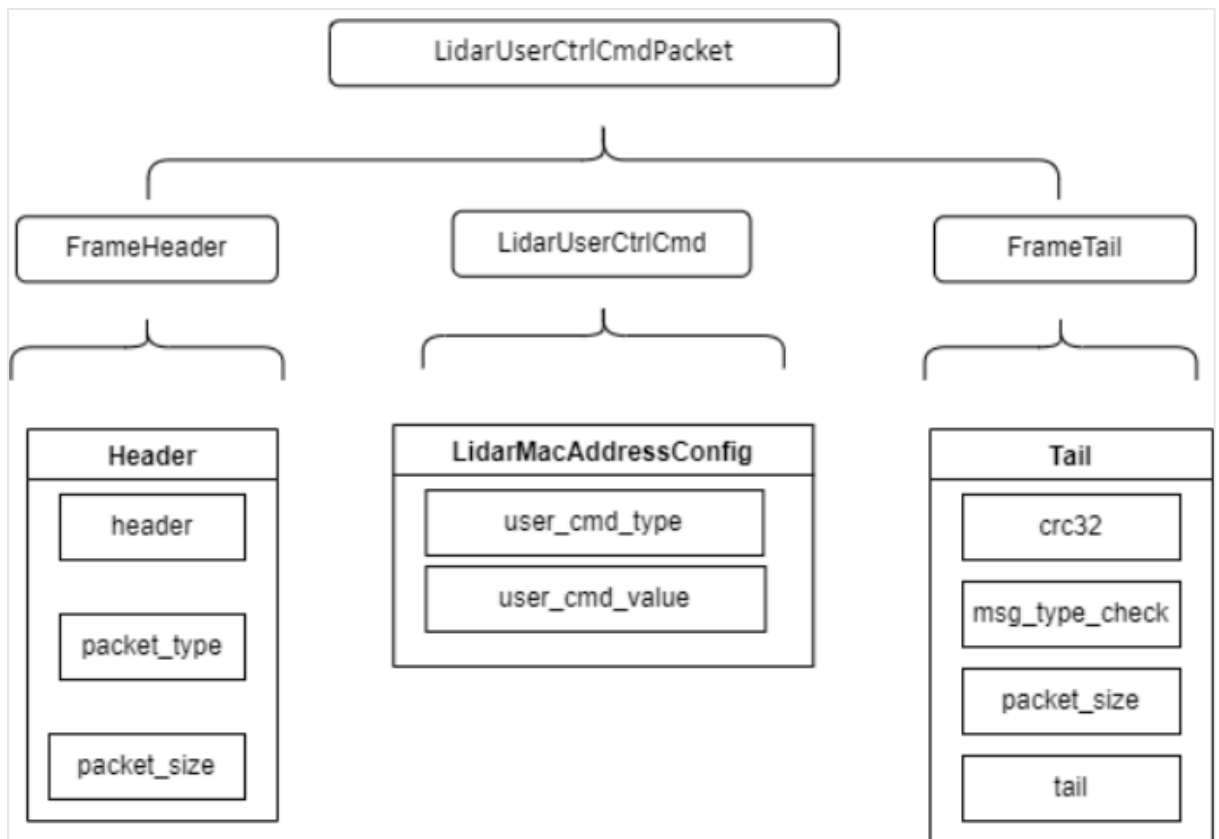
tail[2]: check frame tail, 0x00, 0xFF.

LidarUserCtrlCmdPacket

LidarUserCtrlCmdPacket can send control commands to the radar to perform corresponding state and mode switching. The total length of the data packet is 32 bytes, with 12 bytes of Header, 8 bytes of command field, and 12 bytes of Tail. The commands currently provided are as follows:

- System reset
- StandBy mode
- Get version number
- Delay measurement

- Configuration mode reset



1. Frame header

The frame header is 12 bytes in total, which is used to identify the data location and type.

Header: The 4 bytes of the field are defined as 0x55, 0xAA, 0x05, 0x0A, which are used to identify the start position of the data.

packet_type: The field is used to indicate the type of the current data packet.

Packet_size: The field is used to indicate the size of the current data packet.

2. User command field

user_cmd_type: User command type setting field.

USER_CMD_RESET_TYPE radar reset command.

USER_CMD_STANDBY_TYPE Lidar switch StandBy mode command.

USER_CMD_VERSION_GET Get radar version number command.

USER_CMD_LATENCY_TYPE Request to calculate communication delay command.

USER_CMD_CONFIG_RESET Configuration mode restore default value command, power off to save the default working mode.

user_cmd_value : User command value setting field.

No need to set.

3. Frame tail

crc32: CRC32 checksum value of LidarPointData.

Reserve: uint32 reserve.

reserve [2]: uint8 reserve.

tail[2]: Check frame tail, 0x00, 0xFF.